

AMAN KUMAR SINGH

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Summary

Aspiring AI Engineer and Data Science enthusiast with a strong foundation in Python, SQL, Machine Learning, Data Analytics, and Generative AI. Experienced through academic projects and internship work in developing data-driven applications, predictive models, and backend systems using Python and FastAPI. Proficient in Power BI, data visualization, machine learning, and backend development, with growing expertise in LLMs, RAG pipelines, and LangChain. Seeking opportunities in AI Engineering, Data Science, Data Analytics, and Generative AI.

Education

Master of Computer Application

Sharda University, Uttar Pradesh

PG: 2026

CGPA: 9.27 / 10

Bachelor of Computer Application

JIS College of Engineering, West Bengal

UG: 2024

CGPA: 8.14 / 10

Experience

Healthletic Lifestyle

Backend Engineering Intern

Remote

Jun 2026 – Present

- Developed and maintained backend APIs using Node.js, enabling secure and efficient data exchange between services.
- Designed and optimized database operations using MongoDB, improving query performance and application reliability.
- Collaborated with the development team to implement authentication, bug fixes, and new backend features for healthcare-focused applications, while participating in testing, debugging, deployment activities, and version control workflows using Git.

Projects

Heart-AI: Dual-Engine Cardiovascular Diagnostic | Python, Flask, TensorFlow, XGBoost

Mar 2026



- Engineered a dual-engine cardiovascular diagnostic platform utilizing **Flask, XGBoost, and CNN-based deep learning** to concurrently analyze tabular clinical vitals and physical ECG image scans.
- Trained and deployed high-accuracy machine learning models on complex medical datasets, achieving a **92%** prediction accuracy for real-time heart disease risk assessment and classification.
- Designed a complete full-stack analytics dashboard integrating **SQLite, Chart.js, and FPDF** to deliver interactive health visualizations, patient management archiving, and automated hospital-grade PDF reports.

Air Quality Analysis & Forecasting Dashboard | Python, Flask, LSTM

Jan 2026



- Developed a web-based air quality monitoring system integrating real-time environmental data using APIs to track AQI and pollutant levels (PM2.5, PM10, NO₂, CO, O₃).
- Performed data cleaning and exploratory data analysis (EDA) using **Pandas and NumPy** to clean, transform, and analyze air quality datasets.
- Built a **LSTM-based deep learning model** to forecast PM2.5 levels for the next 24 hours, enabling predictive analysis of pollution trends.

Technical Skills

- **Programming Languages:** Python, SQL, Java, JavaScript
- **Data Science & Analytics:** Pandas, NumPy, Scikit-learn, Power BI, Excel, Data Visualization, EDA
- **Machine Learning:** Supervised & Unsupervised Learning, Deep Learning, Time Series Forecasting, Feature Engineering
- **Generative AI:** LangChain, RAG, Hugging Face, ChromaDB, Vector DB, Prompt Engineering, LLM Applications
- **Frameworks & Backend:** FastAPI, Flask, Node.js, Express.js, Spring Boot
- **Databases:** MySQL, PostgreSQL, MongoDB, SQLite
- **Tools & Platforms:** Git, GitHub, Postman, VS Code, Jupyter Notebook

Certifications

- **Data Visualization using Power BI:** Sharda Informatics — May–June 2025 
- **Prompt Engineering:** Sharda Informatics — Jan–April 2025 
- **AI: Transformative Learning:** TechSaksham — 2025 
- **Python Pro Bootcamp:** Udemy — 2024 
- **Mastering in C:** Udemy — 2022 